

Proximal Balancing

Yonghan Jung

We study causal adjustment with observed pretreatment covariates X , a high-dimensional pretreatment proxy $W = (W_1, \dots, W_J)$, and an unobserved confounder. For one prespecified block j , the proposed cross-view construction learns a finite representation $Z_j = \phi_j(X, W_{-j})$ without using the held-out block W_j , and measures treatment imbalance through the conditional distribution of (X, W_j) . We prove an exact bias decomposition and show that exact cross-view balance identifies the population average treatment effect under joint latent exchangeability, held-out-view separation, overlap, and outcome-relevant completeness. When balance is approximate, a uniform outcome-relevant stability condition yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. A finite-class learning analysis and an honest augmented inverse probability weighted estimator give an end-to-end finite-sample bound that separates representation error, hidden-confounding sensitivity, nuisance error, and evaluation-sample variation. The causal interpretation depends on the stated cross-view structural conditions; the existence of one generic correlated proxy is not sufficient by itself.

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1. Introduction

1.1. Related Works

Balancing scores and covariate balancing. The propensity score is the canonical reduction of observed pretreatment covariates: under treatment ignorability given X , conditioning on $e(X) = P(A = 1 | X)$ balances the distribution of X across treatment arms and identifies causal effects (Rosenbaum and Rubin, 1983). Entropy balancing and the covariate-balancing propensity score instead choose weights or treatment models that target balance directly (Hainmueller, 2012; Imai and Ratkovic, 2014). These methods motivate our optimization target, but their usual causal guarantee still requires the measured covariates to suffice for confounding control. Our setting keeps a latent confounder and asks what a held-out proxy view can certify about the bias remaining after a learned reduction of the other pretreatment measurements.

Causal representation learning. Representation-learning methods compress high-dimensional covariates while penalizing imbalance between treated and control representations. For example, counterfactual regression bounds counterfactual prediction error using factual prediction error and an integral probability metric between learned treatment-arm representations (Shalit et al., 2017). Our construction shares the use of a learned low-dimensional representation and a distributional balance penalty. The difference is the cross-view role of the proxy: for one prespecified block j , the learner constructs $Z_j = \phi_j(X, W_{-j})$ without outcomes and evaluates conditional treatment balance using the held-out variables (X, W_j) , even when ignorability given X may fail.

Negative controls and proximal causal inference. Negative-control variables can reveal residual confounding because they should not be causally affected by the treatment or should not causally affect the outcome, while remaining associated with hidden common causes (Lipsitch et al., 2010). Proximal causal inference turns this diagnostic idea into point identification by assigning distinct treatment-inducing and outcome-inducing proxy roles and imposing completeness and bridge conditions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024); semiparametric proximal theory then develops influence functions and efficient estimators for the resulting identifying functionals (Cui et al., 2024). Our result has a different mechanism: a representation-side view forms the adjustment variable, a held-out view measures residual imbalance, and outcome-relevant completeness transfers exact observed balance to equality of the latent outcome averages needed for the population ATE.

Single-proxy identification. Single Proxy Control shows that one negative-control outcome can identify the effect of treatment on the treated under a counterfactual proxy condition and suitable bridge structure (Park et al., 2024). This differs from our population ATE target, from our latent-confounder proxy interpretation, and from our use of an observable balance discrepancy. The comparison is nevertheless central: it makes clear that “one proxy” does not by itself identify a common estimand. Identification depends on what the proxy measures, which conditional independence is imposed, and whether the analysis uses a bridge, an adjustment score, or an independent audit view.

Kernel discrepancies and sensitivity certification. Maximum mean discrepancy is an integral probability metric that equals zero for identical distributions when its kernel is characteristic (Gretton et al., 2012). We use conditional treatment-arm MMDs of (X, W_j) as observable magnitudes, not kernel-test p -values. The new theoretical object is the product of this discrepancy and an outcome-relevant stability constant, which yields a bias certificate and a representation-selection guarantee. The individual ingredients have close precedents in balancing, representation learning,

kernel testing, and proximal inference; the manuscript therefore makes no priority claim before a systematic literature audit, and instead states its contribution through the exact assumptions, estimand, and guarantee proved below.

2. Problem Setting

2.1. Preliminaries

Structural causal models. A structural causal model (SCM) consists of endogenous variables, exogenous disturbances, and structural assignments. Writing the endogenous variables as $V_1, \dots, V_d$, an acyclic SCM has equations

$$V_j = f_j(\text{pa}_j, E_j), \quad j = 1, \dots, d, \quad (1)$$

where pa_j contains the direct causes of V_j and E_j is an exogenous disturbance. The intervention $\text{do}(A = a)$ replaces the structural equation for A by the constant a while leaving the other equations unchanged. The resulting outcome is the potential outcome $Y(a)$. An average causal effect is *identified* when its value is uniquely determined by the observed-data distribution for every SCM in the stated model class (Pearl, 2009).

Proximal causal inference. Ordinary adjustment seeks observed covariates that block all non-causal paths from treatment to outcome. Proximal causal inference instead allows an unobserved confounder and uses proxy measurements of it. Standard proximal formulations assign two different proxy roles and posit a bridge equation whose solution translates an observable conditional moment into the target causal functional (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Completeness or injectivity prevents distinct latent laws from producing the same observable proxy law. These conditions do not follow from the mere existence of a correlated proxy. Our method uses outcome-relevant completeness to interpret exact cross-view balance and a quantitative stability condition when balance is only approximate.

2.2. Problem Setting

We observe independent copies

$$O_i = (X_i, W_i, A_i, Y_i), \quad i = 1, \dots, n, \quad (2)$$

where X is a possibly high-dimensional vector of observed pretreatment covariates, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. The observed pretreatment measurement W is divided into prespecified blocks,

$$W = (W_1, \dots, W_J). \quad (3)$$

Each block may be scalar or vector-valued. A block can represent one repeated measurement, sensor channel, laboratory panel, or prespecified region of a larger measurement. We call W a candidate proxy because some or all blocks may contain information about an unobserved pretreatment variable U that affects treatment selection and the outcome.

For each $a \in \{0, 1\}$, let $Y(a)$ denote the outcome that would be observed if treatment were set to a . The target is the population average treatment effect

$$\tau \triangleq \mathbb{E}\{Y(1) - Y(0)\}. \quad (4)$$

We allow U to remain associated with treatment after conditioning on X . The research question is whether the block structure of W can support a learned adjustment representation that identifies τ under exact cross-view balance and controls residual hidden-confounding bias when balance is approximate. The representation and the assumptions that give its balance a causal meaning are introduced in Section 3.

3. Identification

We now introduce the cross-view representation. Fix one block index $j \in \{1, \dots, J\}$ before analyzing the data. Let

$$W_{-j} \triangleq (W_1, \dots, W_{j-1}, W_{j+1}, \dots, W_J) \quad (5)$$

contain every proxy block except W_j . The excluded block W_j is the held-out proxy block.

Fix a positive integer M_j , write $[M_j] \triangleq \{1, \dots, M_j\}$, and prespecify a class

$$\Phi_j \triangleq \{\phi : \mathcal{X} \times \mathcal{W}_{-j} \longrightarrow [M_j]\}. \quad (6)$$

For a fixed map $\phi_j \in \Phi_j$, define

$$Z_j \triangleq \phi_j(X, W_{-j}). \quad (7)$$

The representation is finite-valued so that balance, overlap, and adjustment can be stated within explicit strata. This choice does not require the hidden variable U to be discrete. A value $z \in [M_j]$ is retained when $P(Z_j = z) > 0$.

Figure 1 displays three causal graphs compatible with the broad pretreatment ordering. In every panel, W_{-j} is the representation-side proxy view, W_j is the held-out proxy view, and Z_j is the deterministic function defined in equation (7). Solid arrows denote causal relations. Dashed arrows into the rectangular Z_j node denote deterministic construction and are not causal arrows. The graphs alone do not imply the cross-view conditional independences imposed below.

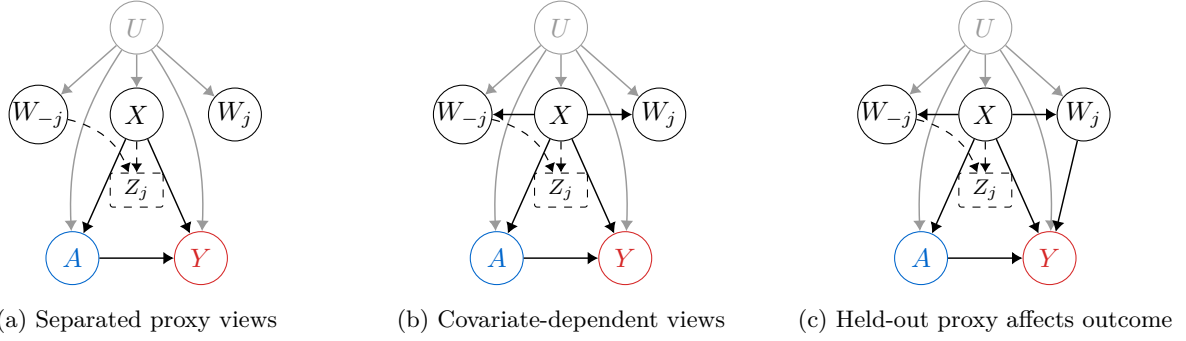


Figure 1: Causal graphs compatible with the cross-view setup. In every panel, W_{-j} is used with X to construct Z_j , while W_j is held out. Solid arrows are causal. Dashed arrows into the rectangular Z_j node show the deterministic construction $Z_j = \phi_j(X, W_{-j})$ and are not causal arrows. Panel (a) lets both proxy views depend on U . Panel (b) additionally lets both views depend on X . Panel (c) shows $W_j \rightarrow Y$; a direct effect $W_{-j} \rightarrow Y$ is also allowed by the stated assumptions but is omitted for legibility. The graphs do not by themselves imply Assumptions 2 and 3.

The identification argument compares the hidden states of treated and untreated units within one retained stratum. For $a \in \{0, 1\}$ and any retained z satisfying $P(A = a, Z_j = z) > 0$, define

$$Q_{a,j,z} \triangleq \mathcal{L}(X, U \mid A = a, Z_j = z). \quad (8)$$

For every measurable set $C \subseteq \mathcal{W}_j$, define the conditional measurement law

$$K_j(C \mid x, u) \triangleq P(W_j \in C \mid X = x, U = u). \quad (9)$$

The law K_j describes how the hidden state (x, u) generates the held-out proxy block.

For any probability distribution Q on (X, U) and measurable set $B \subseteq \mathcal{X} \times \mathcal{W}_j$, define

$$(\mathcal{K}_j Q)(B) \triangleq \int \left[\int 1\{(x, w_j) \in B\} K_j(dw_j \mid x, u) \right] Q(dx, du). \quad (10)$$

Thus, $\mathcal{K}_j Q$ is the distribution of (X, W_j) obtained by first drawing (X, U) from Q and then drawing W_j from its conditional measurement law.

For each treatment level and retained stratum, define

$$\mu_{a,j,z}(x, u) \triangleq \mathbb{E}\{Y(a) \mid X = x, U = u, Z_j = z\}. \quad (11)$$

Let

$$S_{j,z} \triangleq \text{supp}\{\mathcal{L}(X, U \mid Z_j = z)\} \quad (12)$$

be the support of (X, U) within stratum z . Define $\mathcal{Q}_{j,z}$ as the set of all probability distributions Q supported on $S_{j,z}$ for which

$$\int |\mu_{a,j,z}(x, u)| Q(dx, du) < \infty, \quad a \in \{0, 1\}. \quad (13)$$

The two realized arm laws satisfy $Q_{0,j,z}, Q_{1,j,z} \in \mathcal{Q}_{j,z}$. For $Q \in \mathcal{Q}_{j,z}$, write

$$\mathbb{E}_Q\{\mu_{a,j,z}(X, U)\} \triangleq \int \mu_{a,j,z}(x, u) Q(dx, du). \quad (14)$$

3.1. Assumptions

Assumption 1 (Consistency and integrability). *For each $a \in \{0, 1\}$,*

$$Y = Y(A), \quad \mathbb{E}|Y(a)| < \infty. \quad (15)$$

Assumption 2 (Joint latent exchangeability for representation inputs). *For each $a \in \{0, 1\}$,*

$$A \perp\!\!\!\perp (Y(a), W_{-j}) \mid (X, U). \quad (16)$$

Once (X, U) is fixed, treatment contains no additional information about either the potential outcome or the variables used to construct Z_j .

Assumption 3 (Separation of the held-out proxy block).

$$W_j \perp\!\!\!\perp (A, W_{-j}) \mid (X, U). \quad (17)$$

Once (X, U) is fixed, the measurement law of W_j is the same across treatment arms and does not depend on the representation-side proxy blocks.

Assumption 4 (Representation overlap). *For every retained representation value z ,*

$$0 < P(A = 1 \mid Z_j = z) < 1. \quad (18)$$

Both treatment arms therefore occur with positive probability in every retained stratum.

Assumption 5 (Exact outcome-relevant completeness). *For every retained z , every $Q, Q' \in \mathcal{Q}_{j,z}$, and every $a \in \{0, 1\}$,*

$$\mathcal{K}_j Q = \mathcal{K}_j Q' \implies \mathbb{E}_Q\{\mu_{a,j,z}(X, U)\} = \mathbb{E}_{Q'}\{\mu_{a,j,z}(X, U)\}. \quad (19)$$

Two latent distributions that are indistinguishable through (X, W_j) must agree on the mean of each potential outcome. The assumption does not require the two latent distributions themselves to be equal.

3.2. Exact Cross-View Identification

Lemma 1 (Conditioning safety). *Under Assumption 2, for each $a \in \{0, 1\}$,*

$$Y(a) \perp\!\!\!\perp A \mid (X, U, Z_j). \quad (20)$$

Under Assumption 3,

$$W_j \perp\!\!\!\perp (A, Z_j) \mid (X, U), \quad (21)$$

and hence

$$P(W_j \in C \mid X, U, A, Z_j) = K_j(C \mid X, U) \quad (22)$$

for every measurable set C .

Proof. The representation Z_j is a measurable function of (X, W_{-j}) . Assumption 2 therefore implies

$$A \perp\!\!\!\perp (Y(a), Z_j) \mid (X, U). \quad (23)$$

Weak union gives equation (20). Similarly, Assumption 3 and measurability of Z_j imply equation (21). The conditional law in equation (22) then equals the law in equation (9). $\square$

Definition 1 (Adjustment functional). *For the fixed representation Z_j , define*

$$\tau_{Z_j} \triangleq \sum_{z: P(Z_j=z)>0} P(Z_j = z) [\mathbb{E}(Y \mid A = 1, Z_j = z) - \mathbb{E}(Y \mid A = 0, Z_j = z)]. \quad (24)$$

This is an observed-data functional. Its equality with τ has not yet been asserted.

For each retained stratum, write

$$p_{j,z} \triangleq P(Z_j = z), \quad \pi_{j,z} \triangleq P(A = 1 \mid Z_j = z). \quad (25)$$

Theorem 1 (Exact bias decomposition). *Under Assumptions 1, 2, and 4,*

$$\begin{aligned} \tau_{Z_j} - \tau = \sum_z p_{j,z} & \left[(1 - \pi_{j,z}) \left[\mathbb{E}_{Q_{1,j,z}} \{\mu_{1,j,z}\} - \mathbb{E}_{Q_{0,j,z}} \{\mu_{1,j,z}\} \right] \right. \\ & \left. + \pi_{j,z} \left[\mathbb{E}_{Q_{1,j,z}} \{\mu_{0,j,z}\} - \mathbb{E}_{Q_{0,j,z}} \{\mu_{0,j,z}\} \right] \right]. \end{aligned} \quad (26)$$

Proof. Consistency and Lemma 1 give, for $a \in \{0, 1\}$,

$$\mathbb{E}(Y \mid A = a, Z_j = z) = \mathbb{E}_{Q_{a,j,z}} \{\mu_{a,j,z}(X, U)\}. \quad (27)$$

The law of total expectation over treatment gives

$$\begin{aligned}\mathbb{E}\{Y(a) \mid Z_j = z\} &= \pi_{j,z} \mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(X, U)\} \\ &\quad + (1 - \pi_{j,z}) \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(X, U)\}.\end{aligned}\tag{28}$$

Insert equation (27) into Definition 1. Insert equation (28) into the stratum decomposition of τ . Subtracting the two expressions and collecting the coefficients of each latent arm law gives equation (26). $\square$

Theorem 1 isolates the obstacle to adjustment by Z_j . Unbiased adjustment requires equality of two outcome-relevant latent averages within each stratum. Equality of the full latent distributions is sufficient but is not necessary.

Lemma 2 (Cross-view factorization). *Under Assumption 3, for every treatment arm and retained stratum,*

$$\mathcal{L}(X, W_j \mid A = a, Z_j = z) = \mathcal{K}_j Q_{a,j,z}.\tag{29}$$

Proof. Let B be a measurable subset of the joint space of (X, W_j) . Iterated expectation gives

$$\begin{aligned}P\{(X, W_j) \in B \mid A = a, Z_j = z\} \\ = \int P\{(X, W_j) \in B \mid X = x, U = u, A = a, Z_j = z\} Q_{a,j,z}(dx, du).\end{aligned}\tag{30}$$

Lemma 1 replaces the inner conditional law of W_j by $K_j(\cdot \mid x, u)$. The resulting integral is $(\mathcal{K}_j Q_{a,j,z})(B)$ by equation (10). Equality for every measurable B proves the result. $\square$

Definition 2 (Exact cross-view balance). *The fixed representation satisfies exact cross-view balance when*

$$(X, W_j) \perp\!\!\!\perp A \mid Z_j.\tag{31}$$

Under Assumption 4, this condition is equivalent to equality of the two conditional laws of (X, W_j) in every retained stratum.

Exact cross-view balance is a population condition on observed variables for a fixed representation. Lemma 2 converts it into

$$\mathcal{K}_j Q_{1,j,z} = \mathcal{K}_j Q_{0,j,z}\tag{32}$$

for every retained stratum.

Theorem 2 (Exact identification under cross-view balance). *Suppose Assumptions 1 through 5 hold for the fixed block j and fixed representation $Z_j = \phi_j(X, W_{-j})$. If Z_j satisfies exact*

cross-view balance, then

$$\tau_{Z_j} = \tau. \quad (33)$$

Consequently, the population ATE is point identified by the observed functional in equation (24).

Proof. Fix a retained stratum. Exact cross-view balance and Lemma 2 give equation (32). Both latent arm laws belong to $\mathcal{Q}_{j,z}$. Assumption 5 therefore gives, for each $a \in \{0, 1\}$,

$$\mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(X, U)\} = \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(X, U)\}. \quad (34)$$

Both treatment-arm differences in Theorem 1 are zero. Hence $\tau_{Z_j} - \tau = 0$. $\square$

Remark 1 (Identification without full latent balance). *Theorem 2 equates the outcome-relevant latent averages. It does not require $Q_{1,j,z} = Q_{0,j,z}$ and does not assert $(X, U) \perp\!\!\!\perp A \mid Z_j$. This distinction is the reason Assumption 5 is weaker than full mixture injectivity.*

Remark 2 (Attainability). *Theorem 2 applies when the chosen representation attains exact cross-view balance. It does not guarantee that such a map exists in Φ_j . A restricted class may leave positive residual imbalance, which motivates the next subsection.*

3.3. Approximate Cross-View Bias Bound

Define the held-out observed vector

$$V_j \triangleq (X, W_j). \quad (35)$$

Let κ_j be a bounded, measurable, positive-semidefinite kernel on the support of V_j , and let $\mathcal{H}_j$ be its reproducing kernel Hilbert space. For a distribution P of V_j , define its kernel mean embedding

$$m_j(P) \triangleq \int \kappa_j(v, \cdot) P(dv) \in \mathcal{H}_j. \quad (36)$$

For two distributions P and P' , define

$$\text{MMD}_{\kappa_j}(P, P') \triangleq \|m_j(P) - m_j(P')\|_{\mathcal{H}_j}. \quad (37)$$

Assume that κ_j is characteristic, meaning that this discrepancy is zero exactly when $P = P'$.

Definition 3 (Cross-view discrepancy). *For each retained stratum, define*

$$\Delta_{j,z}(\phi_j) \triangleq \text{MMD}_{\kappa_j}\{\mathcal{L}(V_j \mid A = 1, Z_j = z), \mathcal{L}(V_j \mid A = 0, Z_j = z)\}. \quad (38)$$

The aggregate discrepancy is

$$D_j(\phi_j) \triangleq \sum_z p_{j,z} \Delta_{j,z}(\phi_j), \quad (39)$$

where the sum is over retained strata. Both quantities are population functionals of observed variables.

Proposition 1 (Zero discrepancy and exact balance). *Under Assumption 4 and the characteristic-kernel condition,*

$$D_j(\phi_j) = 0 \iff (X, W_j) \perp\!\!\!\perp A \mid Z_j. \quad (40)$$

Proof. Every summand in equation (39) is nonnegative and every retained stratum has positive mass. Thus, the aggregate is zero exactly when every stratum-specific MMD is zero. Characteristic-ness converts each zero MMD into equality of the two conditional laws of (X, W_j) . Under overlap, these equalities over all retained strata are equivalent to conditional independence. $\square$

Lemma 2 also gives

$$\Delta_{j,z}(\phi_j) = \text{MMD}_{\kappa_j}(\mathcal{K}_j Q_{1,j,z}, \mathcal{K}_j Q_{0,j,z}). \quad (41)$$

Assumption 6 (Uniform outcome-relevant stability). *There is a finite constant Γ_j such that, for every retained z , every $a \in \{0, 1\}$, and every $Q, Q' \in \mathcal{Q}_{j,z}$,*

$$\begin{aligned} & |\mathbb{E}_Q\{\mu_{a,j,z}(X, U)\} - \mathbb{E}_{Q'}\{\mu_{a,j,z}(X, U)\}| \\ & \leq \Gamma_j \text{MMD}_{\kappa_j}(\mathcal{K}_j Q, \mathcal{K}_j Q'). \end{aligned} \quad (42)$$

The constant Γ_j converts an observed cross-view discrepancy into an upper bound on an outcome-relevant latent difference. It is a structural sensitivity parameter and is not generally identified from the observed distribution.

Theorem 3 (Approximate cross-view bias bound). *Suppose Assumptions 1 through 4 and Assumption 6 hold for the fixed representation. Then*

$$|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j). \quad (43)$$

Proof. For each treatment level and retained stratum, define

$$\delta_{a,j,z} \triangleq \mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(X, U)\} - \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(X, U)\}. \quad (44)$$

Theorem 1 and the triangle inequality give

$$|\tau_{Z_j} - \tau| \leq \sum_z p_{j,z} [(1 - \pi_{j,z})|\delta_{1,j,z}| + \pi_{j,z}|\delta_{0,j,z}|]. \quad (45)$$

Assumption 6 and equation (41) imply

$$|\delta_{a,j,z}| \leq \Gamma_j \Delta_{j,z}(\phi_j). \quad (46)$$

Substitution, the identity $(1 - \pi_{j,z}) + \pi_{j,z} = 1$, and Definition 3 give equation (43). $\square$

Remark 3 (Exact completeness and quantitative stability). *Assumption 5 preserves equality at zero discrepancy. Assumption 6 controls nonzero discrepancies and therefore implies exact outcome-relevant completeness. The converse need not hold because exact equality can be preserved even when no finite Lipschitz constant Γ_j exists.*

Remark 4 (Sensitivity interpretation). *For a scientifically specified Γ_j , Theorem 3 gives the population sensitivity interval*

$$\tau \in [\tau_{Z_j} - \Gamma_j D_j(\phi_j), \tau_{Z_j} + \Gamma_j D_j(\phi_j)]. \quad (47)$$

A small $D_j(\phi_j)$ alone does not guarantee small causal bias. The causal quantity is the product $\Gamma_j D_j(\phi_j)$.

Remark 5 (Uniformity across candidate representations). *For representation learning, let $\Gamma_j(\phi)$ be a stability budget for candidate ϕ . A common finite budget*

$$\sup_{\phi \in \Phi_j} \Gamma_j(\phi) \leq \bar{\Gamma}_j < \infty \quad (48)$$

is required to compare candidates by discrepancy alone. Without this condition, a representation can have a small observed discrepancy and an arbitrarily large stability constant.

4. Representation Learning and AIPW Estimation

The identification results concern one fixed representation. We now select a representation from data and estimate the adjustment functional that it induces. The analysis uses one prespecified held-out block j throughout.

4.1. Representation Learning

Partition the observed sample into three mutually disjoint subsets,

$$\mathcal{I}_D, \quad \mathcal{I}_N, \quad \mathcal{I}_E, \quad (49)$$

with respective sizes n_D , n_N , and n_E . The discrepancy-learning sample $\mathcal{I}_D$ selects a representation. The nuisance-learning sample $\mathcal{I}_N$ estimates the outcome regressions and propensity used by AIPW. The evaluation sample $\mathcal{I}_E$ calculates the final estimate. Conditional on the preceding samples, the objects evaluated on the next sample are fixed.

For the finite-sample learning result, fix a finite candidate library

$$\mathcal{F}_j \subseteq \Phi_j, \quad N_j \triangleq |\mathcal{F}_j| < \infty. \quad (50)$$

For $\phi \in \mathcal{F}_j$, define

$$Z_j^\phi \triangleq \phi(X, W_{-j}) \quad (51)$$

and let $\mathcal{R}_j(\phi)$ be its set of positive-mass strata. Assume there are constants $p_{\min} > 0$ and $\eta \in (0, 1/2]$ such that every candidate and every $z \in \mathcal{R}_j(\phi)$ satisfy

$$P(Z_j^\phi = z) \geq p_{\min}, \quad (52)$$

$$\eta \leq P(A = 1 \mid Z_j^\phi = z) \leq 1 - \eta. \quad (53)$$

Assume also that the bounded kernel satisfies

$$\sup_v \sqrt{\kappa_j(v, v)} \leq \rho_j < \infty. \quad (54)$$

The primitive cross-view Assumptions 2 and 3 do not depend on the selected candidate. Representation overlap and outcome-relevant stability must hold for every candidate-specific representation. Their stability budgets are required to obey the common bound in equation (48).

Definition 4 (Empirical cross-view discrepancy). *For observation $i \in \mathcal{I}_D$ and candidate $\phi \in \mathcal{F}_j$, write*

$$Z_{j,i}^\phi \triangleq \phi(X_i, W_{-j,i}), \quad V_{j,i} \triangleq (X_i, W_{j,i}). \quad (55)$$

For treatment arm a and stratum z , define

$$N_{a,z}^\phi \triangleq \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\}, \quad (56)$$

$$N_z^\phi \triangleq N_{0,z}^\phi + N_{1,z}^\phi, \quad (57)$$

$$\hat{p}_{j,z}(\phi) \triangleq \frac{N_z^\phi}{n_D}. \quad (58)$$

When $N_{a,z}^\phi > 0$, define the empirical kernel mean

$$\hat{m}_{a,j,z}(\phi) \triangleq \frac{1}{N_{a,z}^\phi} \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\} \kappa_j(V_{j,i}, \cdot). \quad (59)$$

When both treatment-arm cells are nonempty, define

$$\hat{\Delta}_{j,z}(\phi) \triangleq \|\hat{m}_{1,j,z}(\phi) - \hat{m}_{0,j,z}(\phi)\|_{\mathcal{H}_j} \quad (60)$$

and

$$\hat{D}_j(\phi) \triangleq \sum_{z: N_z^\phi > 0} \hat{p}_{j,z}(\phi) \hat{\Delta}_{j,z}(\phi). \quad (61)$$

If an occupied empirical stratum contains only one treatment arm, set $\widehat{D}_j(\phi) = +\infty$.

Proposition 2 (Uniform deviation over the finite library). *Let $\delta_D \in (0, 1)$ and suppose*

$$n_D \geq \frac{8}{p_{\min}\eta} \log \left(\frac{8M_j N_j}{\delta_D} \right). \quad (62)$$

Define

$$e_{j,n_D}(\delta_D) \triangleq \rho_j \left[2 + \sqrt{2 \log \left(\frac{4M_j N_j}{\delta_D} \right)} \right] \sqrt{\frac{2}{n_D p_{\min}\eta}} \quad (63)$$

and

$$\begin{aligned} r_{j,n_D}(\delta_D) &\triangleq 2e_{j,n_D}(\delta_D) \\ &\quad + 2\rho_j \sqrt{\frac{2 \left[M_j \log 2 + \log \left(\frac{4N_j}{\delta_D} \right) \right]}{n_D}}. \end{aligned} \quad (64)$$

Under equations (52) through (54), with probability at least $1 - \delta_D$,

$$\sup_{\phi \in \mathcal{F}_j} |\widehat{D}_j(\phi) - D_j(\phi)| \leq r_{j,n_D}(\delta_D). \quad (65)$$

Proof. Equation (62) and a multiplicative Chernoff bound imply that every candidate treatment-by-stratum cell contains at least $n_D p_{\min}\eta/2$ observations except on an event of probability at most $\delta_D/4$. Conditional on the cell assignments, bounded Hilbert-space mean concentration and a union bound over at most $2M_j N_j$ cells give

$$|\widehat{\Delta}_{j,z}(\phi) - \Delta_{j,z}(\phi)| \leq 2e_{j,n_D}(\delta_D) \quad (66)$$

simultaneously, with failure probability at most $\delta_D/2$. A multinomial L_1 concentration bound gives

$$\sum_z |\widehat{p}_{j,z}(\phi) - p_{j,z}(\phi)| \leq \sqrt{\frac{2 \left[M_j \log 2 + \log (4N_j/\delta_D) \right]}{n_D}} \quad (67)$$

simultaneously over candidates, with failure probability at most $\delta_D/4$. On the intersection of the three events, use $0 \leq \Delta_{j,z}(\phi) \leq 2\rho_j$ and the triangle inequality to obtain equation (65). The three failure probabilities sum to δ_D . $\square$

Definition 5 (Approximate empirical minimizer). *Let $\varepsilon_{\text{opt},j} \geq 0$. The representation learner returns $\widehat{\phi}_j \in \mathcal{F}_j$ satisfying*

$$\widehat{D}_j(\widehat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} \widehat{D}_j(\phi) + \varepsilon_{\text{opt},j}. \quad (68)$$

The learned representation is

$$\widehat{Z}_j \triangleq \widehat{\phi}_j(X, W_{-j}). \quad (69)$$

The tolerance records optimization error and is zero for an exact empirical minimizer.

Theorem 4 (Representation-learning oracle inequality). *Suppose the conditions of Proposition 2 hold. Suppose Assumptions 1 through 3 hold for block j , and suppose representation overlap and uniform outcome-relevant stability hold for every candidate with the common budget $\bar{\Gamma}_j$. Then, with probability at least $1 - \delta_D$,*

$$D_j(\widehat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}. \quad (70)$$

On the same event,

$$|\tau_{\widehat{Z}_j} - \tau| \leq \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right]. \quad (71)$$

Proof. On the uniform-deviation event,

$$D_j(\widehat{\phi}_j) \leq \widehat{D}_j(\widehat{\phi}_j) + r_{j,n_D}(\delta_D) \quad (72)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} \widehat{D}_j(\phi) + \varepsilon_{\text{opt},j} + r_{j,n_D}(\delta_D) \quad (73)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}. \quad (74)$$

This proves equation (70). Conditional on $\mathcal{I}_D$, the selected map is fixed. Theorem 3 and the common stability budget give $|\tau_{\widehat{Z}_j} - \tau| \leq \bar{\Gamma}_j D_j(\widehat{\phi}_j)$. Substitution proves equation (71). $\square$

Remark 6 (Practical learner outside the finite-library theorem). *The theorem covers hard, finite-valued representations in a finite prespecified library. A neural learner using random features, soft assignments, and Adam requires additional bounds for neural-class complexity, feature approximation, soft-to-hard transfer, and optimization error. These guarantees are not supplied by Theorem 4.*

4.2. AIPW Estimation

Conditional on $\mathcal{I}_D$, the map $\widehat{\phi}_j$ is fixed. For a fresh observation, define the conditional outcome means and representation propensity

$$m_{a,j}(z) \triangleq \mathbb{E}(Y \mid A = a, \widehat{Z}_j = z), \quad (75)$$

$$e_j(z) \triangleq P(A = 1 \mid \widehat{Z}_j = z). \quad (76)$$

For the observed adjustment functional $\tau_{\hat{Z}_j}$, the efficient influence function in the nonparametric model for $(\hat{Z}_j, A, Y)$ is

$$\begin{aligned} \psi_j(O) &\triangleq m_{1,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j) \\ &\quad + \frac{A\{Y - m_{1,j}(\hat{Z}_j)\}}{e_j(\hat{Z}_j)} - \frac{(1 - A)\{Y - m_{0,j}(\hat{Z}_j)\}}{1 - e_j(\hat{Z}_j)} - \tau_{\hat{Z}_j}. \end{aligned} \quad (77)$$

Use $\mathcal{I}_N$ to estimate the nuisance functions. Denote the estimates by $\hat{m}_{0,j}$, $\hat{m}_{1,j}$, and $\hat{e}_j$.

Definition 6 (Honest AIPW estimator). *For $i \in \mathcal{I}_E$, define $\hat{Z}_{j,i} = \hat{\phi}_j(X_i, W_{-j,i})$ and*

$$\begin{aligned} S_j(O_i) &\triangleq \hat{m}_{1,j}(\hat{Z}_{j,i}) - \hat{m}_{0,j}(\hat{Z}_{j,i}) \\ &\quad + \frac{A_i\{Y_i - \hat{m}_{1,j}(\hat{Z}_{j,i})\}}{\hat{e}_j(\hat{Z}_{j,i})} \\ &\quad - \frac{(1 - A_i)\{Y_i - \hat{m}_{0,j}(\hat{Z}_{j,i})\}}{1 - \hat{e}_j(\hat{Z}_{j,i})}. \end{aligned} \quad (78)$$

The honest AIPW estimator is

$$\boxed{\hat{\tau}_j^{\text{AIPW}} \triangleq \frac{1}{n_E} \sum_{i \in \mathcal{I}_E} S_j(O_i).} \quad (79)$$

Proposition 3 (Exact signed nuisance remainder). *Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$,*

$$\begin{aligned} &\mathbb{E}\{S_j(O)\} - \tau_{\hat{Z}_j} \\ &= \mathbb{E}\left[\{\hat{e}_j(\hat{Z}_j) - e_j(\hat{Z}_j)\} \left\{ \frac{\hat{m}_{1,j}(\hat{Z}_j) - m_{1,j}(\hat{Z}_j)}{\hat{e}_j(\hat{Z}_j)} \right. \right. \\ &\quad \left. \left. + \frac{\hat{m}_{0,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j)}{1 - \hat{e}_j(\hat{Z}_j)} \right\} \right]. \end{aligned} \quad (80)$$

Proof. Fix $\hat{Z}_j = z$. Taking the conditional expectation of the score and using equation (76) gives

$$\begin{aligned} \mathbb{E}\{S_j(O) \mid \hat{Z}_j = z\} &= \hat{m}_{1,j}(z) - \hat{m}_{0,j}(z) \\ &\quad + \frac{e_j(z)\{m_{1,j}(z) - \hat{m}_{1,j}(z)\}}{\hat{e}_j(z)} \\ &\quad - \frac{\{1 - e_j(z)\}\{m_{0,j}(z) - \hat{m}_{0,j}(z)\}}{1 - \hat{e}_j(z)}. \end{aligned} \quad (81)$$

Subtract $m_{1,j}(z) - m_{0,j}(z)$ and collect the two nuisance errors. Averaging the resulting identity over $\hat{Z}_j$ proves equation (80). $\square$

The remainder is zero if the propensity estimator is correct. It is also zero if both conditional outcome estimators are correct.

Theorem 5 (Conditional finite-sample AIPW error). *Suppose $|Y| \leq B$ almost surely. Clip the nuisance estimates so that*

$$-B \leq \widehat{m}_{a,j}(z) \leq B, \quad \eta \leq \widehat{e}_j(z) \leq 1 - \eta. \quad (82)$$

For a function g of $\widehat{Z}_j$, define

$$\|g\|_{2,\widehat{Z}_j} \triangleq \left[\mathbb{E}\{g(\widehat{Z}_j)^2 \mid \mathcal{I}_D, \mathcal{I}_N\} \right]^{1/2}. \quad (83)$$

After the two learning samples are observed, define

$$q_{e,j} \triangleq \|\widehat{e}_j - e_j\|_{2,\widehat{Z}_j}, \quad (84)$$

$$q_{a,j} \triangleq \|\widehat{m}_{a,j} - m_{a,j}\|_{2,\widehat{Z}_j}, \quad a \in \{0, 1\}. \quad (85)$$

For $\delta_E \in (0, 1)$, define

$$\begin{aligned} \varepsilon_{\text{AIPW},j} \triangleq & 2B \left(1 + \frac{1}{\eta}\right) \sqrt{\frac{2 \log(2/\delta_E)}{n_E}} \\ & + \frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \end{aligned} \quad (86)$$

Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$, with probability at least $1 - \delta_E$ over $\mathcal{I}_E$,

$$\boxed{|\widehat{\tau}_j^{\text{AIPW}} - \tau_{\widehat{Z}_j}| \leq \varepsilon_{\text{AIPW},j}.} \quad (87)$$

Proof. The boundedness and clipping conditions imply

$$|S_j(O)| \leq 2B \left(1 + \frac{1}{\eta}\right). \quad (88)$$

Conditional on the two learning samples, Hoeffding's inequality bounds the difference between the evaluation average and its conditional expectation by the first term in equation (86). Proposition 3, Cauchy–Schwarz, and the propensity clipping bound control the absolute conditional remainder by

$$\frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \quad (89)$$

The triangle inequality proves equation (87). $\square$

4.3. End-to-End Finite-Sample Bound

Theorem 6 (End-to-end sensitivity bound). *Suppose the conditions of Proposition 2, Theorem 4, and Theorem 5 hold. Then, jointly over the three independent samples, with probability*

at least $1 - \delta_D - \delta_E$,

$$|\hat{\tau}_j^{\text{AIPW}} - \tau| \leq \varepsilon_{\text{AIPW},j} + \bar{\Gamma}_j \left[\hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D) \right]. \quad (90)$$

It also satisfies the oracle form

$$\begin{aligned} |\hat{\tau}_j^{\text{AIPW}} - \tau| &\leq \varepsilon_{\text{AIPW},j} \\ &\quad + \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right]. \end{aligned} \quad (91)$$

Proof. The triangle inequality gives

$$|\hat{\tau}_j^{\text{AIPW}} - \tau| \leq |\hat{\tau}_j^{\text{AIPW}} - \tau_{\hat{Z}_j}| + |\tau_{\hat{Z}_j} - \tau|. \quad (92)$$

Theorem 5 bounds the first term. Theorem 3 and uniform candidate stability bound the second term by $\bar{\Gamma}_j D_j(\hat{\phi}_j)$. Proposition 2 gives

$$D_j(\hat{\phi}_j) \leq \hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D), \quad (93)$$

which proves equation (90). Equation (70) proves the oracle form. A union bound over the discrepancy-learning and evaluation events gives the stated joint probability. $\square$

Corollary 1 (Consistency). *Consider a sequence of candidate libraries, sample splits, nuisance estimators, optimization tolerances, and stability budgets satisfying Theorem 6. If*

$$\delta_{D,n} + \delta_{E,n} \longrightarrow 0, \quad (94)$$

$$\frac{\log(2/\delta_{E,n})}{n_E} \longrightarrow 0, \quad (95)$$

$$q_{e,j,n}(q_{0,j,n} + q_{1,j,n}) \xrightarrow{p} 0, \quad (96)$$

and

$$\bar{\Gamma}_{j,n} \left[\inf_{\phi \in \mathcal{F}_{j,n}} D_j(\phi) + 2r_{j,n_D}(\delta_{D,n}) + \varepsilon_{\text{opt},j,n} \right] \xrightarrow{p} 0, \quad (97)$$

then

$$\hat{\tau}_{j,n}^{\text{AIPW}} \xrightarrow{p} \tau. \quad (98)$$

Proof. The first three conditions make the evaluation-sample and nuisance-product components of $\varepsilon_{\text{AIPW},j}$ converge to zero. Equation (97) makes the stability-weighted representation term vanish. The failure probability of Theorem 6 also converges to zero, which proves convergence in probability. $\square$

Remark 7 (Interpretation and observability). *Theorem 6 separates the final error into an AIPW estimation term and a hidden-confounding sensitivity term. The stability budget $\bar{\Gamma}_j$ and the nuisance errors $q_{e,j}$, $q_{0,j}$, and $q_{1,j}$ are not generally observed. The result becomes an operational numerical certificate after these quantities are bounded by additional theory, external information, or a declared sensitivity analysis.*

Remark 8 (Scope). *The theorem uses one prespecified held-out block, hard finite strata, a finite candidate library, and an honest three-way split. It does not establish a neural-class rate, a soft-to-hard approximation bound, an Adam optimization guarantee, or a cross-fitting theorem. Experimental aggregation across several held-out orientations is also outside this fixed- j result.*

5. Experiments

5.1. Questions, Comparators, and Reporting Rules

The experiments answer three separate questions. First, does proxy-supervised representation adjustment reduce ATE error relative to adjustment using X alone? Second, do the exact-assumption designs recover the known balancing structure? Third, does the method show a useful descriptive signal in an observational benchmark with a noisy randomized reference? We report mean absolute error, root mean squared error, paired wins, held-out discrepancy, overlap diagnostics, and Monte Carlo uncertainty. Oracle procedures that observe U or the true balancing score are simulation references, not implementable competitors. The theory in Sections 3 and 4 applies to one prespecified held-out block at a time. Experimental aggregation across several orientations is reported as an empirical extension and is not presented as a separate identification theorem.

5.2. Honest Approximate SCMs

The primary approximate study fixes the causal order $U \rightarrow (X, W)$, $(X, U) \rightarrow A$, and $(A, X, U) \rightarrow Y$. Both SCMs use $\dim(X) = 20$ and $\dim(W) = 10$, with twenty independently generated data sets and nested prefixes $n \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$. SCM1 uses linear proxy measurements and a nonlinear outcome component; SCM2 changes the structural equations to nonlinear transformations while preserving the same causal order and an informative monotone proxy channel. The proposed cross-mask method learns a representation from X and all but one proxy coordinate, audits balance with the held-out coordinate, and averages over masks.

At $n = 1000$, the mean absolute errors of proposed/ X -adjustment/full- (X, W) /oracle- (X, U) were 0.169/0.601/0.310/0.130 in SCM1 and 0.112/0.440/0.188/0.107 in SCM2. At $n = 10000$, the corresponding proposed errors were 0.199 and 0.134, versus 0.588 and 0.413 for X adjustment. These are finite-sample simulation results, not a proof of the audit-view identification theorem. The self-conditioned heuristic remained substantially biased, which is consistent with the self-conditioning warning in Section 3.

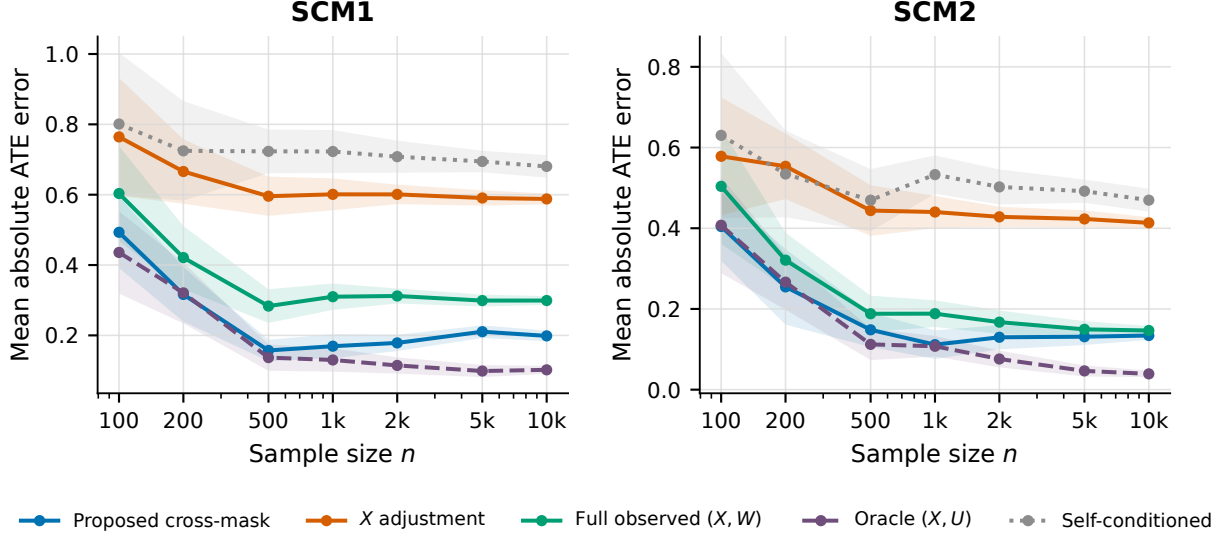


Figure 2: Honest approximate SCMs. Curves show mean absolute ATE error across twenty independently generated data sets; bands are approximate 95% Monte Carlo intervals for the mean. Prefixes are nested within a data set, so uncertainty is computed separately at each n and is not pooled across sample sizes. The proposed cross-mask learner uses one proxy coordinate for held-out auditing and averages across masks.

5.3. Exact-Assumption Tracks A and B

Track A uses a finite balancing score, $\dim(X) = 20$, nine representation-side proxy coordinates, and one independent audit coordinate. The DGP enforces strict population overlap, exact decoding of the discrete confounding state from the representation-side view, and an injective Gaussian audit channel. At $n = 10000$ over twenty replications, mean absolute error was 0.175 for the learned audit representation, 0.433 for X adjustment, 0.186 for direct full-observed adjustment, and 0.033 for the oracle balancing score. The primary comparison with X passed, while the prespecified secondary superiority criterion against full-observed adjustment failed.

Track B uses a continuous exact balancing score and a linear representation class matched to that score. At $n = 10000$, mean absolute error was 0.0748 for the learned representation, 0.1583 for fitted X adjustment, and 0.1342 for fitted full-observed adjustment. The learned method beat X in 20/20 and full-observed adjustment in 19/20 paired replications. Its mean rank-recovery Spearman correlation was 0.915. The held-out conditional-dependence diagnostic used finite random features and is a surrogate, not an exact population KCI test; the design also covers one prespecified scalar- U SCM family.

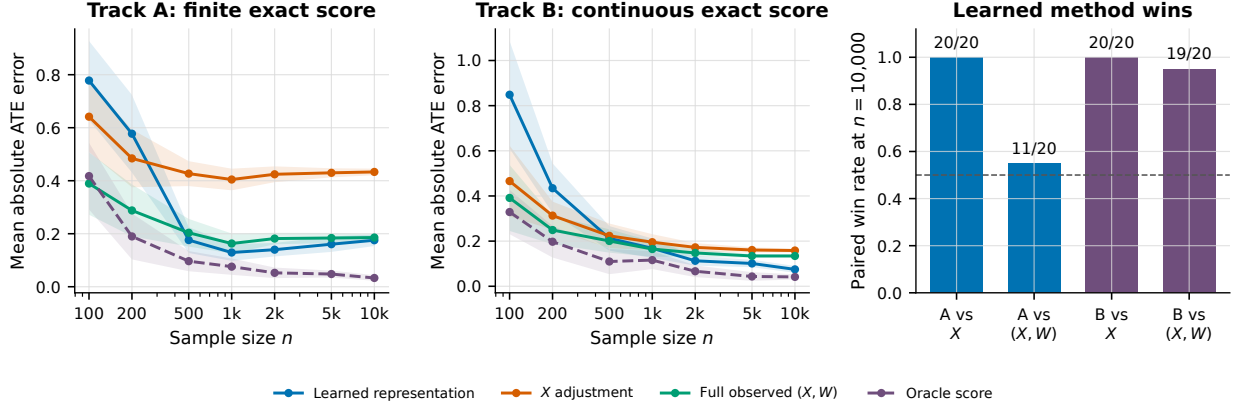


Figure 3: Exact-assumption experiments. The first two panels report mean absolute ATE error with approximate 95% Monte Carlo intervals over twenty replications. The third panel reports paired win rates at $n = 10,000$, where a win means that the learned representation has smaller absolute error than the indicated comparator. Track A satisfies the primary comparison with X but not the prespecified secondary superiority criterion against full observed adjustment. Track B records rank-recovery Spearman correlation 0.915 and held-out finite-RFF conditional-dependence surrogate 0.00691.

5.4. WSC Observational Benchmark

The WSC study forms an observational subset of 1,105 participants whose randomized mathematics assignment agrees with pretreatment preference, and targets the population ATE in the original 2,200 randomized participants. The full randomized difference in means is 0.760 with standard error 0.151 and serves only as a noisy benchmark. Across ten algorithmic split repetitions on this one fixed data set, the unconstrained cross-mask estimate averaged 1.302 with split standard deviation 0.065 and absolute benchmark gap 0.541, compared with gaps 2.690 for the naive observational contrast, 1.704 for direct X adjustment, and 0.672 for direct (X, W) adjustment. The unconstrained cross-mask method is the proposed WSC method shown in Figure 4. Its practical-overlap gate passed 0/10 repetitions. The benchmark therefore supplies descriptive evidence only, not a validated causal-performance claim, causal ground truth, or a test of proxy validity.

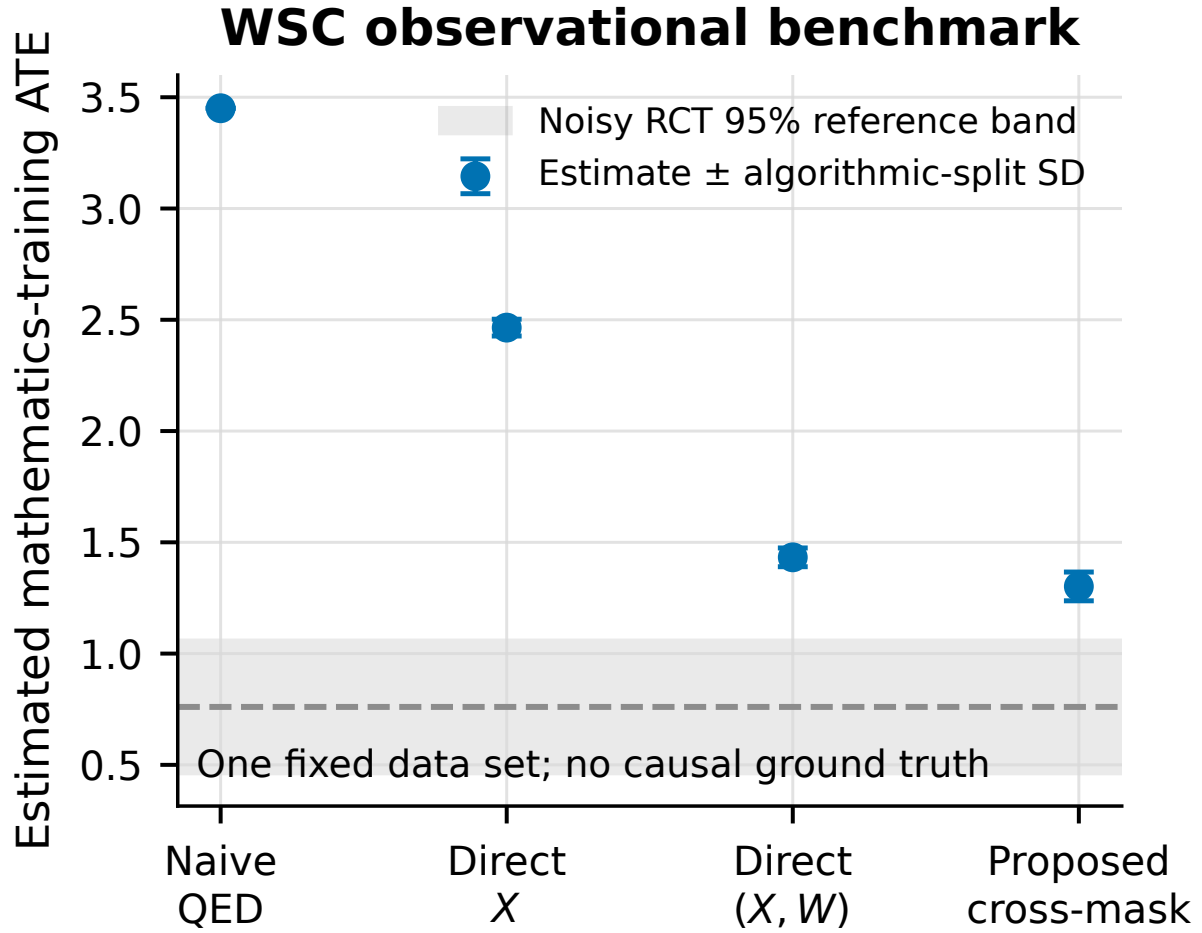


Figure 4: WSC observational benchmark. Estimates are shown relative to the noisy randomized reference. Error bars for fitted methods are standard deviations across ten algorithmic splits of one fixed data set, not Monte Carlo uncertainty. The benchmark has no causal ground truth and reports only the unconstrained cross-mask learner as the proposed method.

6. Discussion and Limitations

6.1. What the Method Delivers

For one prespecified block j , the canonical method uses W_{-j} to construct an outcome-free representation and uses the held-out block W_j to measure the residual imbalance of (X, W_j) across treatment arms within that representation. Under uniform outcome-relevant stability, the product $\Gamma_j D_j(\phi_j)$ is a sensitivity certificate for hidden-confounding bias. This is stronger than reporting a conditional-independence test alone because it targets a causal error scale, but weaker than point identification when $D_j(\phi_j) > 0$ because Γ_j is not generally identified from the observed distribution.

6.2. Testable and Structural Components

Held-out joint balance, empirical overlap, and minimum stratum mass are observable properties. Joint latent exchangeability, separation of the held-out proxy block, outcome-relevant completeness, and outcome-relevant stability are structural assumptions about the full-data law. Exact balance is therefore an observable population condition that can only be assessed up to sampling error, while its causal interpretation still requires structural assumptions. A large held-out discrepancy can falsify exact balance; a small test statistic cannot prove exact balance, completeness, or stability.

6.3. Attainability and Realized Bias

The exact identification theorem is conditional on the existence of a finite representation $Z_j = \phi_j(X, W_{-j})$ that attains exact cross-view balance. The theorem does not guarantee such attainability in a restricted representation class. In the approximate regime, reducing $D_j(\phi_j)$ reduces the certified worst-case bias under a common stability budget, but it need not reduce the realized bias when candidate representations have different stability constants or when the structural stability condition fails.

6.4. Learning and Inference Gaps

The current finite-sample theorem covers enumerable representation classes and fixed-fold evaluation. The neural experiments use soft assignments and covariance-weighted surrogate objectives that are not algebraically identical to the canonical MMD aggregation. A neural-class generalization bound, a soft-to-hard guarantee, and valid joint inference after adaptive representation selection remain open. These gaps prevent the manuscript from describing every neural result as a direct verification of the theory, even when its empirical error is lower.

6.5. Audit Views and Practical Scope

Cross-view learning uses one prespecified proxy block for held-out evaluation and the remaining high-dimensional measurement for representation learning. This design blocks literal self-copying of the evaluated block. It does not by itself establish separation among proxy views, so point identification still needs the structural assumptions in Section 3. The experiments repeat the procedure

across several held-out orientations and aggregate their estimates as an empirical extension; the fixed- j theory applies separately to each orientation. The WSC experiment further shows that an encouraging benchmark gap can coexist with a failed practical-overlap rule. Real-data evidence must therefore report overlap, held-out discrepancies, and benchmark uncertainty without treating any one diagnostic as ground truth.

6.6. Scope of the Contribution

The defensible contribution is the combination of a cross-view proxy-supervised representation, an observable conditional discrepancy, an exact identification theorem, a stability-indexed approximate guarantee, and an honest finite-sample estimation analysis. The paper does not claim that one generic correlated proxy point-identifies the ATE, that every smaller discrepancy has smaller realized bias without a common stability budget, or that the current neural optimizer reaches the population optimum. A systematic literature audit and broader external validation are required before any firstness or universality claim.

7. Conclusion

Single proxy balancing asks what a structured high-dimensional pretreatment proxy can reveal about hidden-confounding bias after learning a low-dimensional adjustment representation. For one prespecified held-out block j , the cross-view construction learns $Z_j = \phi_j(X, W_{-j})$ and evaluates treatment balance through (X, W_j) . Exact cross-view balance identifies the ATE under the stated structural assumptions, while approximate balance yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. Empirical discrepancy minimization and honest AIPW estimation give an end-to-end bound that separates learning, hidden-confounding, nuisance, and evaluation errors. The current simulations support the method in exact and approximate designs while exposing limitations from unattainable balance, weak proxy channels, surrogate mismatch, and practical-overlap loss. Neural-class generalization, soft-to-hard transfer, stability calibration, and external validation remain necessary for a submission-ready paper.

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A. Simulation Details

The existing project-local experiments will be documented by recording each data-generating process, parameter value, sample size, replication count, and random seed policy. Every manuscript figure and table will be linked to the script and machine-readable result from which it was produced.

The simulation audit will separate observed quantities from oracle quantities. In particular, any result that uses U or a direction constructed from U will be labeled as an oracle diagnostic and will not be presented as the output of a learnable procedure.

B. Proof Obligations

The first proof obligation is the conditioning-safety argument for $Z_j = \phi_j(X, W_{-j})$. It must show how joint latent exchangeability and separation of the held-out block remain valid after conditioning on the learned representation.

The second proof obligation is the exact bias decomposition and cross-view factorization. These results must connect the observed adjustment functional, the two treatment-arm latent laws, and the held-out distribution through one common proxy channel.

The third proof obligation is exact identification under outcome-relevant completeness. It must show that exact observed balance makes the two outcome-relevant latent averages equal without asserting equality of the full latent distributions.

The fourth proof obligation is the approximate sensitivity bound. It must show how the stability constant converts each held-out discrepancy into an outcome-relevant latent bound and how the stratum weights aggregate to $\Gamma_j D_j(\phi_j)$.

The fifth proof obligation is the finite-sample chain from uniform discrepancy concentration through empirical representation selection to honest AIPW estimation. Its final inequality must display the representation, stability, nuisance, and evaluation-sample terms separately.